ANTÓNIO JOSÉ FREITAS MACHADO DÓRDIO GOMES

MECHANICAL ENGINEER — STRUCTURAL ANALYSIS & COMPOSITE MATERIALS

PROFILE

Mechanical engineer combining industry experience in mechanical design, integration, prototyping and manufacturing with a strong background in structural analysis and composite materials. My current work includes Ansys static and modal analysis alongside hands-on component development. Previously, I modelled low-velocity impact damage in Glass/PPS laminates at Politecnico di Milano and designed, simulated, manufactured and tested bonded CFRP suspension components for Formula Student at FEUP. My analysis experience covers static, fatigue, impact and non-linear behaviour in Abaqus, with particular emphasis on adhesive joints, damage modelling and experimental validation.

TECHNICAL FOCUS

- MECHANICAL DESIGN & INTEGRATION
- PROTOTYPING & MANUFACTURING
- FINITE ELEMENT ANALYSIS
- ABAQUS · ANSYS
- COMPOSITE STRUCTURES
- IMPACT & DAMAGE MECHANICS
- BONDED JOINTS & ADHESIVES
- EXPERIMENTAL VALIDATION
- STRUCTURAL FATIGUE
- NON-LINEAR MECHANICS
- CAD · FUSION 360
- PYTHON / MATLAB

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Electric Motors

-
Certifications
Awards &
Skills

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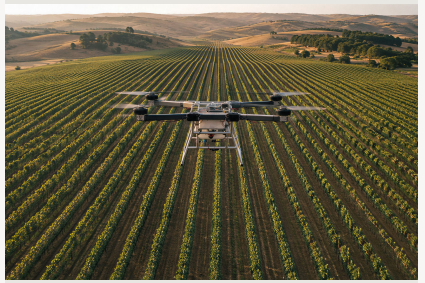
PROJECT 01 · JUN 2026 - PRESENT

Mechanical Design Engineer

UAVision

Mechanical integration, sheet-metal design, prototype development and structural analysis across aerial, ground and counter-UAS platforms.

PRODUCTS	Veteras · MPAV-S · UCDL Z10 · ELANUS
DESIGN & ANALYSIS	Fusion 360 · Ansys · Sheet metal · FDM
MANUFACTURING	Laser cutting · Press brake operation
KEY RESULT	23% lower total jammer assembly weight



VETERAS



MPAV-S



UCDL Z10



ELANUS

PRODUCT IMAGES © UAVISION
Veteras / MPAV-S / UCDL Z10 / ELANUS

Mechanical Design Engineer

Jun 2026 - Present · Fusion 360 CAD · Sheet metal · Hands-on production · FDM

PRODUCT CONTEXT

Veteras - All-electric heavy-lift VTOL with up to 200 kg payload.

MPAV-S - Multipurpose, modular UGV for logistics and casualty evacuation.

UCDL Z10 - Modular, gyrostabilised anti-UAS jammer.

ENGINEERING CONTRIBUTIONS

- **Integrate electronics** into Veteras, MPAV-S and UCDL Z10 through mechanical component design.
- Design **sheet-metal parts in Fusion 360** and personally **operate laser cutting and press brake machines** for in-house production.
- **FDM-print parts for electronics integration**, selecting filaments to suit each application.
- **Develop prototypes and improve existing products**, primarily Veteras and UCDL Z10.

UCDL Z10 · WEIGHT REDUCTION



UCDL Z10 · OFFICIAL PRODUCT IMAGE © UAVISION

23% REDUCTION IN TOTAL JAMMER ASSEMBLY WEIGHT

Replaced machined components with sheet-metal parts in the **UCDL Z10**, reducing **total jammer assembly weight by 23%**. Designed and manufactured the replacement parts in-house.

PRODUCT SOURCES & IMAGES © UAVISION: uavision.com/veteras uavision.com/mpav-s uavision.com/ucdl-z10

Mechanical Design Engineer

Jun 2026 - Present · ELANUS launcher integration · Ansys static and modal analysis

ELANUS · PRODUCT CONTEXT

ELANUS is UAVision's lightweight, tube-deployed loitering munition. The complete system participated in the European Defence Agency's **Sentinel Strike Challenge**.

MY CONTRIBUTION · LAUNCHER INTEGRATION

- Contributed to **CAD design, manufacturing and assembly** for the **ELANUS launcher**.
- Helped resolve **mechanical integration problems** during manufacturing and assembly.

ANSYS · HELICOPTER SUPPORT ANALYSIS

- Performed **static and modal analyses** of a **helicopter battery and gimbal support** in Ansys.
- Assessed **six directional load cases** (upward, downward, forward, rearward, left and right) per applicable **ASTM F3813-26** requirements.
- Compared natural frequencies with **experimentally validated helicopter excitation frequencies** to assess resonance risk.

ELANUS · OFFICIAL PRODUCT IMAGE



ELANUS · PRODUCT IMAGE © UAVISION

1 of 5

FINALISTS FROM 140+
EUROPEAN ENTRIES

Sentinel Strike Challenge · European Defence Agency
ELANUS was selected to progress to Stage 2 and participated in the EU OPEX26 operational experimentation campaign at Santa Margarida, Portugal, in 2026.

PRODUCT & IMAGE: UAVISION · PROGRAMME: EUROPEAN DEFENCE AGENCY

uavision.com/elanus

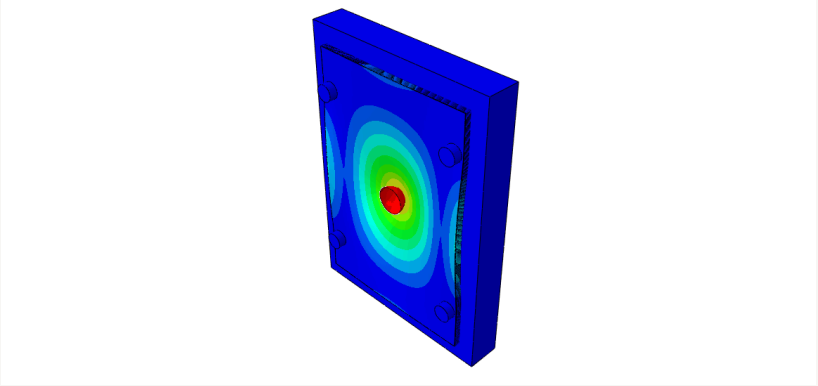
EDA · SENTINEL STRIKE CHALLENGE

PROJECT 02 · SEP 2025 – MAR 2026

Impact Damage Modelling in Glass/PPS Thermoplastic Composite Laminates

Graduate Researcher — Department of Aerospace Science and Technology
Politecnico di Milano, Italy

MATERIAL	Woven Glass/PPS thermoplastic composite laminates
SOLVER	Abaqus/Explicit · Hashin + CZM damage formulation
VALIDATION	Drop-weight impact tests · Ultrasonic C-scan
KEY RESULT	Delamination area predicted within ~15% of C-scan data



FINITE ELEMENT SIMULATION OF IMPACT-INDUCED DEFORMATION IN THE GLASS/PPS LAMINATE MODEL (ABAQUS/EXPLICIT)

MASTER'S THESIS

Low-Velocity Impact Modelling of Glass/PPS Laminates

Graduate Researcher · Sep 2025 – Mar 2026

OBJECTIVE

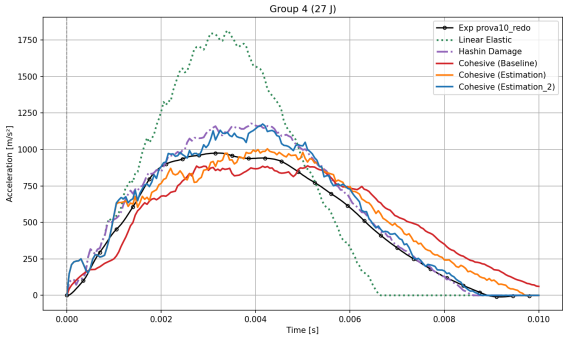
Develop and validate an Abaqus/Explicit finite element model for low-velocity impact in woven Glass/PPS thermoplastic laminates, capable of reproducing the global impact response and the main damage mechanisms observed experimentally.

METHODOLOGY

- Implemented **intralaminar damage modelling** in Abaqus/Explicit using the Hashin 2D criterion
- Introduced **cohesive interface elements** to represent interlaminar delamination
- Developed a **bi-phasic shell–solid modelling strategy** to investigate matrix-dominated deformation and permanent indentation beneath the impactor
- Reproduced the **drop-weight impact configuration** through calibrated boundary conditions, contact definitions and impact loading
- Compared numerical predictions against **experimental impact response** and **ultrasonic C-scan damage measurements**

VALIDATION APPROACH

- Compared predicted **delamination footprints** with ultrasonic **C-scan** measurements
- Correlated numerical **acceleration–time response** with experimental impact data
- Assessed the capability of different modelling strategies to reproduce **back-face cracking**, **delamination**, and **local indentation behaviour**



IMPACT RESPONSE CORRELATION

Acceleration–time comparison between experiment and FE models

KEY RESULTS

~15%
DELAMINATION AREA PREDICTION ERROR VS ULTRASONIC C-SCAN

Bi-phasic
IMPROVED REPRESENTATION OF LOCAL INDENTATION BEHAVIOUR

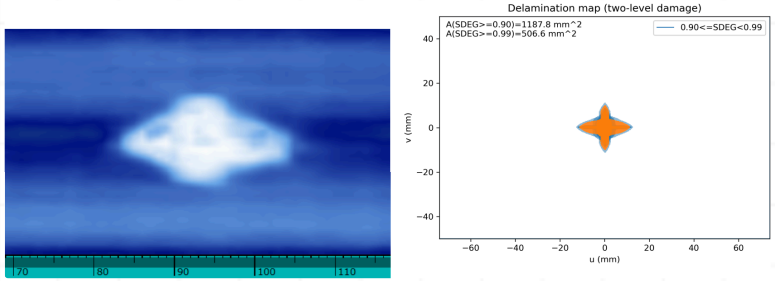
ENGINEERING CONTRIBUTIONS

- Identified the relative roles of **intralaminar fracture**, **interlaminar delamination**, and **matrix-dominated deformation** in the impact response
- Demonstrated that **intralaminar damage modelling** provided the best compromise between accuracy and computational cost for reproducing the global response and back-face crack morphology
- Showed that **cohesive modelling** was necessary to relate numerical predictions to **ultrasonic delamination measurements**
- Used the **bi-phasic approach** to investigate permanent indentation mechanisms not captured by shell-only laminate models

TOOLS & METHODS

- Abaqus/Explicit · Python post-processing
- Ultrasonic C-scan inspection · Drop-weight impact testing
- Finite element damage modelling · Experimental correlation

MODEL VALIDATION



DELAMINATION FOOTPRINT COMPARISON

Ultrasonic C-scan vs cohesive damage prediction

FORMULA STUDENT FEUP · 2023 – 2025

Electric Formula Racing Prototype Development

Composite suspension, bonded joints and transmission engineering
Formula Student FEUP

ROLES	Wishbones & Vibrations Lead · Transmission Engineer
FOCUS	CFRP structures · Adhesive bonded joints · Fatigue · Lightweight drivetrain design
COMPETITIONS	FS Germany · FS Portugal · FS United Kingdom · FS Netherlands · FS Switzerland
HIGHLIGHTS	Design, Cost & Manufacturing and Business Plan awards across 2024–2025



FSFEUP ELECTRIC PROTOTYPE DEVELOPED FOR THE 2025 COMPETITION SEASON

PROJECT 03 · SEP 2024 – AUG 2025

CFRP Suspension Wishbones with Bonded Aluminium Connectors

Wishbones & Vibrations Lead · Formula Student FEUP
Electric Formula Racing Car

MATERIALS	CFRP tubes · 7075-T6 aluminium connectors · structural adhesive
SCOPE	25 suspension load cases · ≈40 bonded joint specimens tested
SIMULATION	Abaqus/Standard · Cohesive Zone Modelling (CZM)
KEY ACHIEVEMENT	Bi-adhesive bonding solution validated to 80°C operating temperature



FORMULA STUDENT REAR SUSPENSION SHOWING CFRP WISHBONES AND ALUMINIUM HUB ASSEMBLY

CFRP Suspension Wishbones with Bonded 7075-T6 Aluminium Connectors

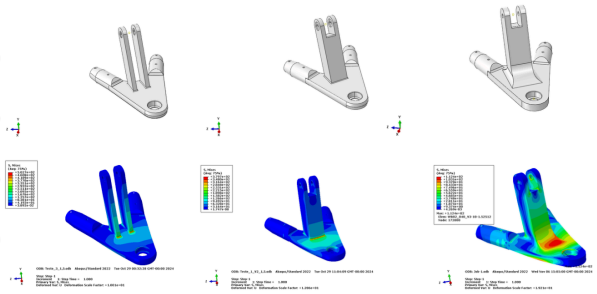
Led a 2-person sub-team · Structural simulation, bonded-joint validation, manufacturing and experimental testing

OBJECTIVE

Manufacture, assemble and validate lightweight CFRP suspension wishbones with bonded 7075-T6 aluminium end connectors for the Formula Student electric race car, ensuring structural performance across suspension load cases and experimental validation of the bonded-joint concept up to 80°C.

STRUCTURAL ANALYSIS

- Performed **Abaqus/Standard simulations of aluminium connectors** across **25 suspension load cases**
- Modelled the adhesive layer using **Cohesive Zone Modelling (CZM)** to evaluate joint failure load and damage evolution
- Correlated simulation predictions with tensile test data from **representative bonded tube–connector specimens**
- Evaluated the team's **bi-adhesive joint concept** through FEM correlation and experimental testing, focusing on **tensile strength**, **repeatability** and **failure mode**



FE ANALYSIS OF 7075-T6 ALUMINIUM CONNECTOR UNDER SUSPENSION LOAD CASES

MANUFACTURING & TESTING

- Manufactured and tested **≈40 representative CFRP tube–connector specimens** in tensile loading to failure
- Assembled and integrated bonded wishbone prototypes, troubleshooting bonding and manufacturing issues during implementation
- Validated the bonded-joint concept experimentally from **room temperature to 80°C**
- Compared observed fracture patterns and failure loads with **numerical predictions**

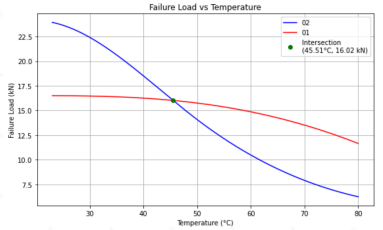
KEY RESULTS

25
SUSPENSION LOAD CASES EVALUATED IN FE SIMULATION

80°C
BONDED JOINT VALIDATED AT OPERATING TEMPERATURE

≈40
BONDED JOINT SPECIMENS MANUFACTURED AND TESTED

5
TEMPERATURE LEVELS TESTED FROM RT TO 80°C

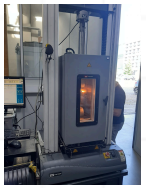


FAILURE LOAD VS TEMPERATURE COMPARISON BETWEEN JOINT SOLUTIONS

MANUFACTURING & TESTING SETUP



MANUFACTURING JIG AND ASSEMBLED WISHBONES



TENSILE TESTING WITH ENVIRONMENTAL CHAMBER

ENGINEERING CONTRIBUTIONS

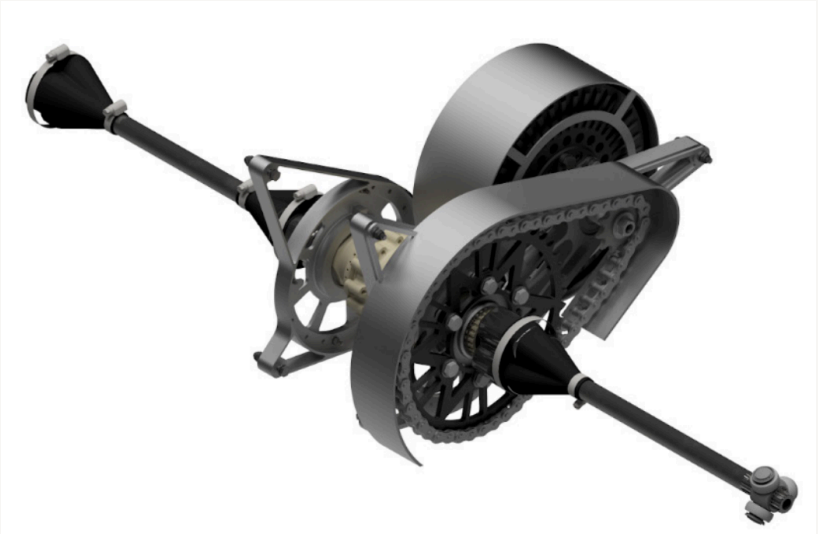
- Continued validation of the team's **bi-adhesive wishbone joint concept** through simulation and experimental testing
- Established correlation between **cohesive FE modelling** and **experimental bonded-joint behaviour**
- Supported suspension subsystem implementation through **manufacturing**, **assembly** and **troubleshooting**

PROJECT 04 · SEP 2023 – SEP 2024

Drivetrain & Transmission Design

Transmission Engineer — Formula Student FEUP
Electric Formula Racing Car FSFEUP-02

COMPONENTS	Tripod housing · Driveshaft · Halfshafts · Sprocket adaptor
ANALYSIS	Static FEM · Safe-life fatigue · Topology optimisation
MASS SAVED	4 kg drivetrain reduction vs previous prototype
TRIPOD HOUSING	35% weight reduction: 1.407 kg → 0.920 kg



FSFEUP-02 DRIVETRAIN ASSEMBLY — DIFFERENTIAL, TRIPOD HOUSING, HALFSHAFTS AND CHAIN DRIVE

Drivetrain Development: Tripod Housing Optimisation, Spline Verification & Fatigue Analysis

Transmission Engineer · Structural simulation, drivetrain integration, fatigue assessment and interface validation

SCOPE

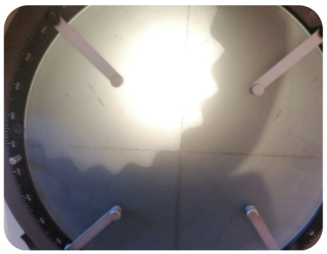
Developed and validated key drivetrain components for the FSFEUP-02 electric prototype, with emphasis on reducing subsystem mass, accommodating a new lower-cost automotive tripod joint, and ensuring structural integrity through FEM, spline verification and safe-life fatigue design.

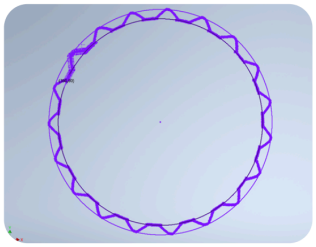
STRUCTURAL DEVELOPMENT

- Performed **static FEM analysis** of the tripod housing and sprocket adaptor in Abaqus, assessing stresses and safety factors under drivetrain torque loads
- Iterated the **tripod housing** through three design stages, moving from a two-material concept to a fully steel optimised solution
- Validated the final tripod housing under **880 Nm torsional loading**, achieving a verified structural safety factor after mass reduction
- Executed **safe-life fatigue analysis** of the halfshafts using LTS torque spectra and S-N based assessment methods

TRIPOD JOINT & INTERFACE VALIDATION

- Adopted a **street-car tripod joint** solution to **reduce cost by 800€** and simplify local manufacturability compared with custom FSAE tripod interfaces
- Performed **spline reverse engineering** using an involute projector and CAD reconstruction of the FIAT 127 tripod interface
- Verified that the new tripod interface follows **DIN 5480 N 21 × 1 × 19**, enabling a more standardised spline definition for manufacturing
- Used the **LTS torque spectrum** as input for safe-life halfshaft dimensioning and fatigue validation

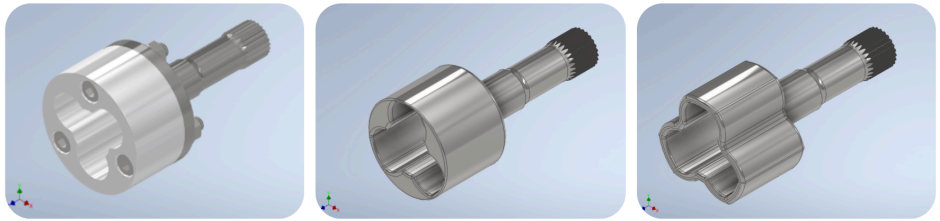




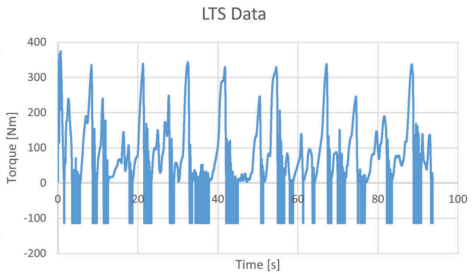
SPLINE RECONSTRUCTION AND DIN 5480 VERIFICATION OF THE FIAT 127 TRIPOD INTERFACE

WEIGHT REDUCTION RESULTS

COMPONENT	INITIAL	FINAL	REDUCTION
Tripod Housing	1.407 kg	**0.920 kg**	**−35%**
Sprocket Adaptor	0.251 kg	**0.123 kg**	**−51%**
Drivetrain Total	—	**−4 kg**	**vs prev. prototype**



TRIPOD HOUSING EVOLUTION FROM INITIAL CONCEPT TO FINAL LIGHTWEIGHT DESIGN



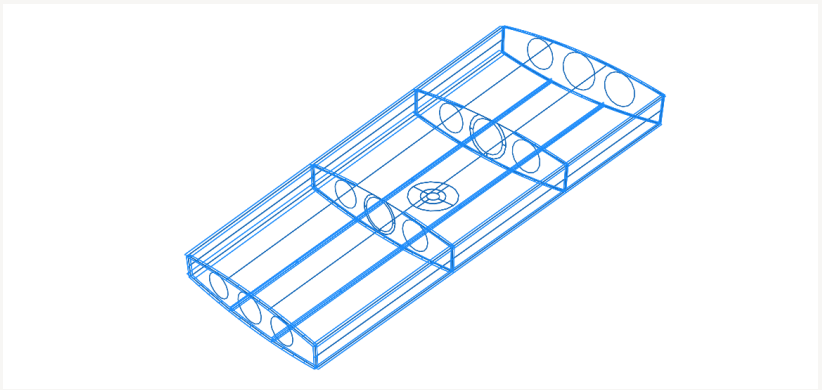
TORQUE SPECTRUM FROM LAP-TIME SIMULATION USED FOR HALFSHAFT FATIGUE ASSESSMENT

ACADEMIC PROJECTS · MSC MECHANICAL ENGINEERING

Structural & Aerospace Engineering

Finite Element Analysis · Structural Mechanics · Failure Mechanics
Selected projects in aerospace structures and advanced structural analysis

TOPICS	Aircraft wing-box analysis · Composite materials · Nonlinear structural behaviour
METHODS	Finite Element Analysis · Buckling · Fatigue analysis · Failure criteria
TOOLS	Abaqus · MATLAB · Python · Analytical structural methods
PROJECTS	Wing-box structural analysis · Aircraft materials & joints · Nonlinear mechanics



FINITE ELEMENT ANALYSIS OF AN AIRCRAFT WING-BOX STRUCTURE

Aircraft Wing-Box Static Structural Analysis

Abaqus FEM · MATLAB · Excel · Aerospace structural verification

OBJECTIVE

Evaluate the structural behaviour of an aircraft wing-box under limit load conditions and determine certification safety margins for the main structural members including skins, spars and ribs.

METHODOLOGY

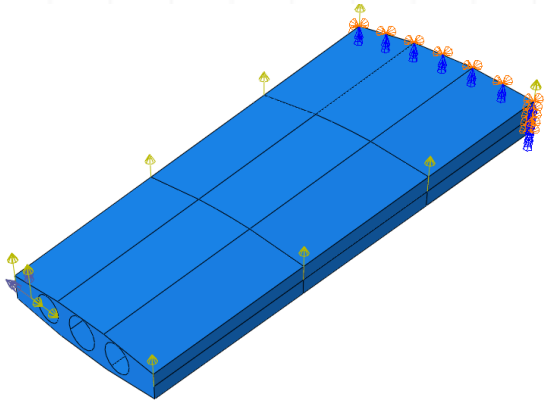
- Developed a detailed **Abaqus finite element model** of the wing-box including skins, spars, ribs and stringers
- Applied representative **aerodynamic loads and boundary conditions** to simulate wing structural response
- Extracted stress fields, internal forces and displacement distributions from FEM results
- Post-processed results in **MATLAB and Excel** to compute margins of safety for multiple failure modes
- Evaluated structural certification criteria including **tension, global buckling, local buckling and crippling**

KEY RESULTS

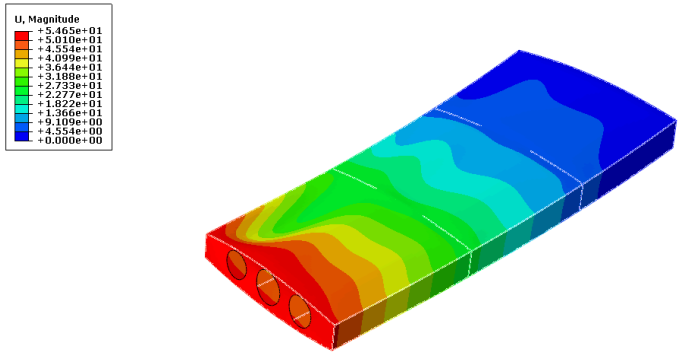
COMPONENT	FAILURE MODE	MS
Spar S1	Global Buckling	**−0.60**
Spar S2	Global Buckling	**−0.59**
Skin P1	Normal Crippling	**−0.31**
Rib N2	Tension	10.52

ENGINEERING INSIGHT

- Negative safety margins in **spars S1 and S2** indicate susceptibility to global buckling under compressive loading
- **Skin panel P1** shows insufficient normal crippling resistance, suggesting the need for structural reinforcement or geometry modification
- The analysis highlights the importance of **buckling-driven design constraints** in lightweight aerospace wing structures



FINITE ELEMENT MODEL OF THE WING-BOX WITH APPLIED LOADS AND BOUNDARY CONDITIONS



GLOBAL DISPLACEMENT FIELD OF THE WING-BOX UNDER LIMIT LOAD CONDITIONS

Aircraft Wing-Box Fatigue & Damage Tolerance Analysis

Composite failure criteria · Safe-life fatigue · Crack growth modelling

OBJECTIVE

Assess the fatigue behaviour and damage tolerance of an aircraft wing-box, combining composite ply-level verification of the upper skin with safe-life and inspection-based evaluation of critical metallic components.

METHODOLOGY

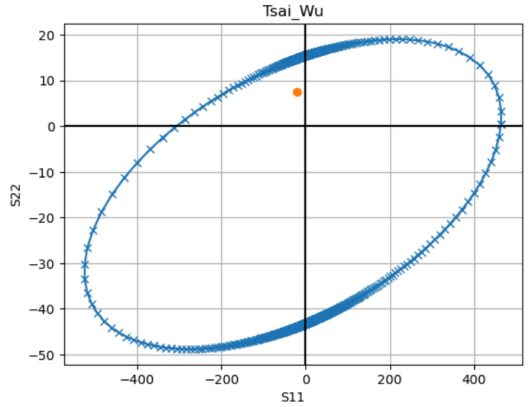
- Modified the wing-box model to include a **[0/+45/−45/90]s CFRP upper skin laminate** and local stress-raising geometric features
- Evaluated composite ply behaviour under two gust load cases using **Maximum Stress, Maximum Strain, Tsai-Hill and Tsai-Wu** criteria
- Computed **stress concentration factors** for the inspection hole and rib N3 central hole
- Derived **fatigue notch factors** using Neuber-based relations
- Performed **safe-life fatigue analysis** for Rib N3 and **damage tolerance analysis** for the lower inspection hole using crack-growth methods

KEY RESULTS

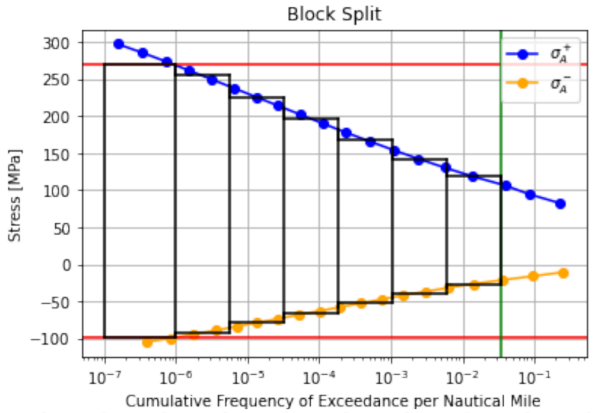
COMPONENT	ASSESSMENT	RESULT
Upper Skin (CFRP)	Infinite Life	**Plies 4 & 5 not certified**
Rib N3	Safe Life	7.38×10^5 FH
Inspection Hole	First Inspection	2.30×10^7 FH
Inspection Hole	Re-inspection	3,026 FH

ENGINEERING INSIGHT

- The **CFRP upper skin satisfies static certification**, but the two 90° plies fall below the required infinite-life fatigue margin under the critical gust case
- **Rib N3 achieves safe-life certification**, demonstrating adequate fatigue resistance under the derived mission spectrum
- The inspection-hole study shows how **damage tolerance methods** can be used to define inspection planning rather than immediate redesign



TSAI-WU FAILURE ENVELOPE USED FOR COMPOSITE PLY STRENGTH ASSESSMENT



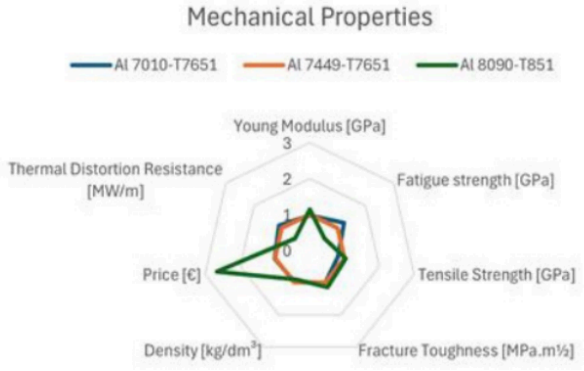
STRESS SPECTRUM AND BLOCK DISCRETIZATION USED FOR FATIGUE LIFE ASSESSMENT

Aircraft Structural Materials — Light Alloys & Composite Laminate Mechanics

Material selection · MATLAB · Classical Laminated Plate Theory · Monte Carlo analysis

1 — LIGHT ALLOYS: AIRBUS A380 WING RIB MATERIAL SELECTION

- Investigated **premature crack formation in Airbus A380 wing rib feet**, associated with the use of Al 7449 and its lower fatigue and thermal distortion resistance for this application
- Performed a **material selection study** using literature review and the **GRANTA EduPack Level 3 Aerospace database** to compare candidate aluminium alloys for wing rib structures
- Analysed the trade-off between **Al 7449 and Al 7010** in terms of fatigue strength, fracture toughness, stiffness and thermal distortion resistance
- Extended the study to alternative alloys from the **2000 and 8000 series**, including Al 2024-T351, Al 2090-T83 and Al 8090-T851
- **Engineering conclusion:** Al 7010 provides the most balanced solution for rib feet, while Al 8090 offers attractive specific stiffness but insufficient fatigue performance for safe-life design

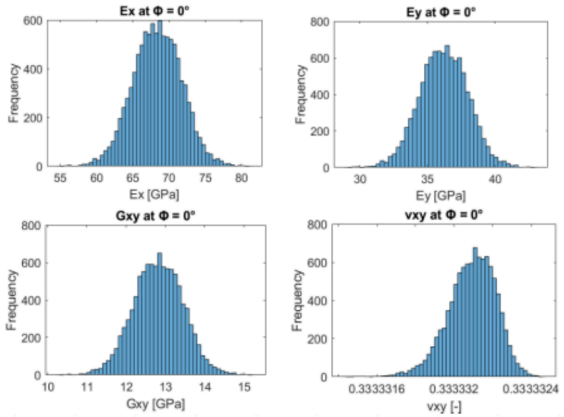


SELECTED MECHANICAL PROPERTIES COMPARISON

ALLOY	Σ_{UTS} [MPA]	FATIGUE [MPA]	K_{IC} [MPA√M]
Al 7449-T7651	**605**	219	29.6
Al 7010-T7651	488	**269**	**29.6**
Al 2024-T351	392	140	36.4
Al 8090-T851	504	142	34.0

2 — COMPOSITE LAMINATE MECHANICS

- Developed **MATLAB routines** to determine the effective elastic properties of AS4/3501-6 CFRP laminates from **micro-mechanics**, using Rule of Mixtures and **Tsai's modulus / invariant approach**
- Applied **Classical Laminated Plate Theory (CLPT)** to multidirectional laminates and **Double-Double laminates [±Φ/±Ψ]**, evaluating the evolution of **E_x, E_y, G_{xy} and ν_{xy}** with ply orientation
- Performed **Monte Carlo sensitivity analyses** to quantify the effect of constituent Young's modulus variability, ply-angle deviations of ±5°, and their combined influence on laminate stiffness
- Compared predicted laminate behaviour across different Φ/Ψ angle combinations, showing the design flexibility of **Double-Double laminate architectures**
- **Engineering insight:** even small manufacturing deviations in ply angles can propagate into measurable variability in laminate engineering constants, reinforcing the importance of **uncertainty quantification** in composite design



3 — Adhesive Bonding of Aluminium–Composite Structures

Experimental testing · Abaqus cohesive modelling · Failure prediction

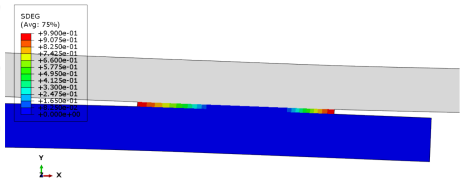
ADHESIVE JOINT INVESTIGATION

- Manufactured and tested **aluminium–CFRP single-lap joints (SLJ)** to investigate bonded-joint structural behaviour.
- Compared two aerospace structural adhesives: **Araldite AV138** (epoxy paste adhesive) and **AF 163-2K** (film adhesive commonly used in aerospace structures).
- Specimens were prepared with controlled adhesive thickness and **25 mm overlap length**, following bonded-joint manufacturing procedures.
- Performed **tensile testing** using a universal testing machine to determine failure load and failure mode.

FINITE ELEMENT MODELLING

- Developed a **2D Abaqus model** to simulate bonded-joint mechanical behaviour.
- Adherends were discretized with **CPE4R elements** and the adhesive layer with **COH2D4 cohesive elements**.
- The cohesive-zone model captured **damage initiation and propagation** in the adhesive layer.
- Simulation results were compared with experiments to **validate failure prediction capability**.

ARALDITE AV138 — ADHESIVE FAILURE



FEM PREDICTION

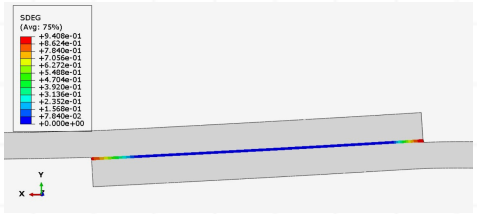


EXPERIMENTAL FAILURE

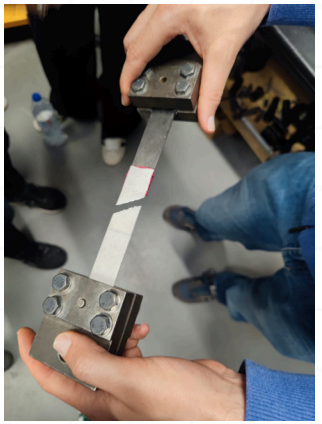
EXPERIMENTAL VS NUMERICAL RESULTS

- **Araldite AV138:** failure load **≈ 5.16 kN**, with cohesive failure in the adhesive layer.
- **AF 163-2K:** failure load **≈ 18.5 kN**, with failure occurring in the **aluminium adherend**.
- Finite element predictions showed good agreement with experiments (**≈ 4% error** for AV138 and **≈ 7.6% error** for AF163-2K).
- Results confirmed the superior structural performance of **AF 163-2K film adhesive** for aluminium–composite bonding.

AF 163-2K — ADHEREND FAILURE



FEM PREDICTION



EXPERIMENTAL FAILURE

Non-Linear Buckling of a Hollow Al 7075-T6 Tube

Abaqus/Standard · Linear Perturbation · Static, Riks · Imperfection Sensitivity

OBJECTIVE

Investigate the buckling behaviour of a thin-walled aluminium tube and compare **linear eigenvalue** and **non-linear FEM** predictions, including the effects of **geometric imperfections** and **material plasticity**.

MODEL DEFINITION

PARAMETER	VALUE
Geometry	Ø 28 mm · t = 2 mm · L = 1000 mm
Material	Al 7075-T6
Element	S8R shell
BCs	Bottom fixed · top pinned + axial load

ANALYSIS WORKFLOW

- Validated FEM against **Euler analytical buckling load**
- Extracted first **buckling eigenmode** using linear perturbation
- Introduced **geometric imperfection** from eigenmode before the Riks step
- Performed **Static, Riks analysis** with and without plasticity

KEY RESULTS

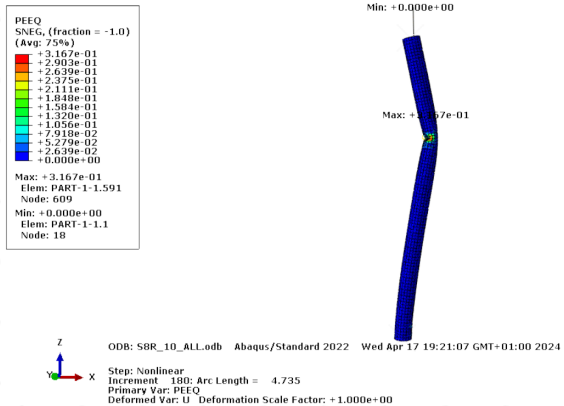
0.67%
ERROR BETWEEN EULER ANALYTICAL SOLUTION AND FEM EIGENVALUE RESULT

181 kN
CRITICAL BUCKLING LOAD FROM LINEAR PERTURBATION ANALYSIS

159 kN
FULLY NON-LINEAR BUCKLING LOAD INCLUDING PLASTICITY

ENGINEERING INSIGHT

- **Imperfection implementation** is essential to activate realistic buckling behaviour in Riks analysis
- Including both **geometric and material non-linearity** reduced the predicted buckling load relative to the idealized model
- Plastic strain localised at the buckle region, demonstrating interaction between **instability and yielding**



PLASTIC STRAIN LOCALISATION IN THE BUCKLE REGION DURING FULLY NON-LINEAR RIKS ANALYSIS

Thermal Modelling of Heat Dissipation in Electric Motor Housings

Product Development Department · Bachelor's Thesis Internship · ANSYS Steady-State Thermal

OBJECTIVE

Evaluate the heat dissipation performance of electric motor housings during the transition from **cast-iron housings** to **steel housings with welded fins**. Develop and validate **ANSYS thermal FEM models** to support design decisions regarding fin geometry and joining configuration.

METHODOLOGY

- Developed **steady-state thermal FEM models in ANSYS** to simulate heat dissipation in motor housings
- Compared **cast-iron housings** with **welded steel fin configurations**
- Performed **parametric analysis of fin geometry**: chamfered vs non-chamfered welded fins
- Conducted **experimental validation** using PT100 sensors, controlled airflow and thermographic imaging
- Used **Python scripts** for heat-transfer calculations and result post-processing

KEY RESULTS

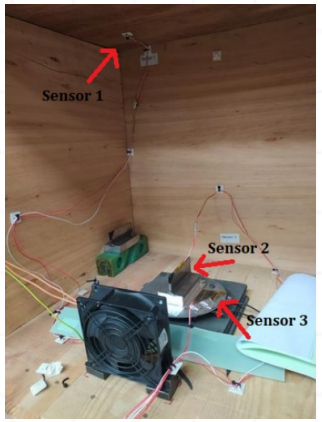
34–40%
HEAT-DISSIPATION LOSS OF CHAMFERED STEEL FINS RELATIVE TO CAST-IRON HOUSING

48–52%
HEAT-DISSIPATION LOSS OF NON-CHAMFERED STEEL FINS RELATIVE TO CAST-IRON HOUSING

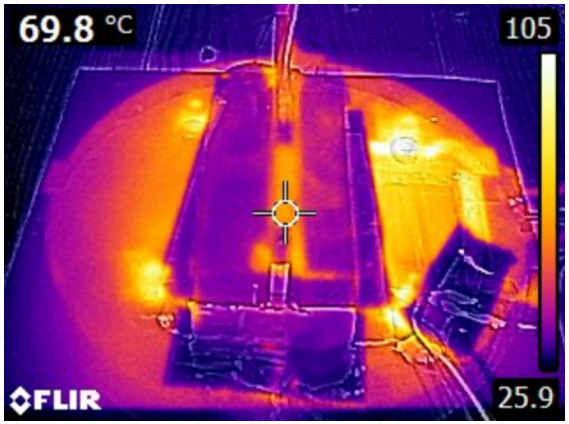
7.7%
MAXIMUM DEVIATION BETWEEN CORRECTED FEM MODEL AND EXPERIMENTAL MEASUREMENTS

ENGINEERING INSIGHTS

- Chamfered steel fins improved thermal contact and clearly outperformed non-chamfered fins
- Thermal contact resistance and radiation effects were essential to match simulation and experiment
- The study combined **thermal simulation, test setup design and experimental validation** in an industrial context



EXPERIMENTAL SETUP WITH PT100 SENSORS AND FORCED AIRFLOW FOR THERMAL VALIDATION



THERMOGRAPHIC MEASUREMENT OF HOUSING TEMPERATURE DISTRIBUTION DURING TESTING

TOOLS USED

- ANSYS Workbench · Python · SolidWorks · Excel · Thermal instrumentation

FEA & SIMULATION

- **Abaqus** — Explicit, Standard, Riks; damage, CZM, topology optimisation
- **ANSYS** — Thermal, static and modal FEM
- **Python** — Scripting, parametric model automation
- **MATLAB** — CLPT, homogenisation, fatigue, data processing

CAD & DESIGN

- **SolidWorks** — CSWA Certified (Aug 2023)
- **Inventor** · **Fusion 360**
- **KISSoft** — Gear and spline analysis

DOCUMENTATION

- LaTeX · Excel · PowerPoint

LANGUAGES

- **Portuguese** — Native
- **English** — C1
- **Spanish** — B1
- **French** — A2

CERTIFICATIONS

SW

CSWA — Certified SolidWorks Associate
Dassault Systèmes · Aug 2023
Credential: C-2JAUPXEG85

ST

Composites Design Workshop XXVII
Prof. Steve Tsai — Stanford University
Aeronautics & Astronautics · Jul 2024

ENGINEERING FOCUS AREAS

- COMPOSITE STRUCTURES & DAMAGE MECHANICS
- FINITE ELEMENT ANALYSIS (ABAQUS/EXPLICIT & STANDARD)
- STRUCTURAL FATIGUE & DAMAGE TOLERANCE
- IMPACT DAMAGE MECHANICS
- ADHESIVE BONDED JOINTS & CZM
- NON-LINEAR STRUCTURAL MECHANICS
- AERONAUTICAL STRUCTURAL DESIGN
- EXPERIMENTAL VALIDATION METHODS

FORMULA STUDENT AWARDS 2025 (EV CLASS)

1st BUSINESS PLAN FS PORTUGAL	**2nd** ENGINEERING DESIGN FS PORTUGAL	**3rd** COST REPORT FS PORTUGAL
3rd COST REPORT FS GERMANY	**9th** BUSINESS PLAN FS GERMANY	**1st** BUSINESS PLAN FS NETHERLANDS

FORMULA STUDENT AWARDS 2024 (CONCEPT CLASS)

1st OVERALL FS SWITZERLAND	**1st** DESIGN & COST FS SWITZERLAND	**2nd** OVERALL FS UK
1st COST REPORT FS UK	**1st** OVERALL FS PORTUGAL	**1st** DESIGN & COST FS PORTUGAL

CONTACT

ENGINEERING PORTFOLIO · 2026

ANTÓNIO GOMES

MECHANICAL ENGINEER
AERONAUTICAL STRUCTURES & COMPOSITES

AREAS OF INTEREST

- Aerospace Structures
- Composite Materials & Structural Simulation
- Motorsport & High-Performance Engineering

Open to graduate engineering roles in structural simulation, composite materials and aerospace structures.

Thank you for your time and consideration.

CONTACT INFORMATION

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EDUCATION

MSc in Mechanical Engineering — Aeronautical Structures & Vehicles
FEUP · 2023 - 2026 · Thesis at Politecnico di Milano

BSc in Mechanical Engineering
FEUP · 2020 – 2023